DE CALCULO INTEGRALI.

Auctore

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CAlculus Integralis inversus est Differentialis: nam modum docet, quo ex proposita quantitate infinitesimali, altera quantitas finita, cuius differentiatione orta est illa, elici debet, quoties hoc fieri potest. Nam non omnis quantitas differentialis integralem algebraicam habet, sed sæpissime transcendentem, quæ per quadraturas tantum curvarum exhiberi potest, per artificia hactenus cognita. Itaque genuina methodus calculi integralis media suppeditare debet, quibus certo cognosci possit: an differentialis aliqua proposita integrabilis sit, an vero secus; &, si primum obtinet, quænam sit eius summa algebraica; sin vero talem non habeat, qualis sit figura simpliciſſima earum, quarum areæ exhibent integralem quæsitam, idque non solum tum, cum data quantitas differentialis unicam indeterminatam cum constantibus involvit, sed tunc quoque, cum duas aut plures indeterminatas cum suis differentialibus complectitur. An vero methodus, quam hoc loco communicabo, ea, quæ requiri dixi, præstet, atque hanc ob causam generalis vocari mereatur, intelligentium iudicio & æquitati id omne diiudicandum relinquo. *M. Mai. 1726.*

PRÆNOTANDA.

1. Æquationes integrandas sic generaliter deinceps

exprimam. 1°. per $du = R^{\lambda} dK$, 2°. $du = R^{\lambda} S^{\mu} dK$. 3° $du = R^{\lambda} S^{\mu} T^{\nu} dK$, &c. & sic infinitum. In quibus R, S, T, &c. denotant quantitates una, duabus, aut pluribus indeterminatis ut libet compositas; elementum vero dK non modo easdem indeterminatas, quas quantitates R, S, T, &c. continent, involvere potest, sed etiam plures, aut subinde etiam secunda differentialia. Litteræ græcæ λ, μ, ν exponentes indefinitas quantitatum R, S, T, designant, casibus particularibus deinceps definiendas.

2. Logarithmos quantitatum quarumlibet per litteram l quantitatibus illis præfixam, usitato more, posthac significabo, nisi aliquando expresse monendum sit alio sensu hanc litteram l adhiberi. Itaque lx, ly, lx^m significant logarithmos quantitatum x, y, & x^m. &c.

3. Per litteram vero $\int$ quantitatibus differentialibus præpositam intelligi volo summam seu integrale quantitatis, cui eadem littera est præfixa. Adeoque $\int R^{\lambda} dK$ denotat integralem vel summam elementi $R^{\lambda} dK$.

Lemma 1.

Integrale fractionis cuiusque $\frac{adx}{x}$, *cuius numerator est multiplum* a *elementi denominatoris, est logarithmus denominatoris* x *ad potestatem* a *evecti. Hoc est* $\int \frac{adx}{x} = lx^a$.

Lemma 2.

Quantitas absoluta, quæ competit seriei cuilibet logarithmorum signis + *vel* − *quomodocunque connexorum, est factum singularum quantitatum singulis logarithmis respondentium exponente affirmativa vel negativa affectarum, prout logarithmi earum signo affirmativo vel negativo præditi fuerint.*

Quan-

Quantitas abſoluta ſeriei $lx+ly-lz$ eſt $x^l y^l z^{-l}$, hoc eſt $\frac{xy}{z}$. Et quantitas abſoluta huius $alR+lM$ eſt $R^a M$.

Ambo hæc lemmata notiora ſunt, quam ut demonſtratione, cuilibet in hiſce vel leviter tantum verſato facilime invenienda, egeant, ſed tanquam fundamenta ſequentibus erant præmittenda.

Theorema 1.

Integralis æquationis $du=R^\lambda dK$, *eſt* $u=MR^{\lambda+1}$, *in qua* M *datur per æquationem hanc* $dK=(\lambda+1)MdR+RdM$.

Hanc æquationem differentialem in ſequentibus *Canonicam* vocabo, quia revera regulam ſuppeditat ſecundum quam æſtimatio litteræ M tuto inveniri poteſt.

Dem. Differentiando æquationem $u=MR^{\lambda+1}$, fit $du=(\lambda+1)MR^\lambda dR+R^{\lambda+1}dM=[(\lambda+1)MdR+RdM]R^\lambda$ vel propter *(hyp)* $dK=(\lambda+1)MdR+RdM$, habetur $du=R^\lambda dK$. Q. E. D.

Theorema 2.

Integralis æquationis $du=R^\lambda S^\mu dK$, *eſt* $u=MR^{\lambda+1}S^{\mu+1}$, *& æquatio* Canonica $dK=(\lambda+1)MSdR+(\mu+1)MRdS+RSdM$.

Dem Aequatio $u=MR^{\lambda+1}S^{\mu+1}$ differentiata præbet $du=[(\lambda+1)MSdR+(\mu+1)MRdS+RSdM]R^\lambda S^\mu$, hoc eſt propter $dK=(\lambda+1)MSdR+(\mu+1)MRdS+RSdM$, fit $du=R^\lambda S^\mu dK$. Q. E. D.

Theorema 3.

Aequatio $du=R^\lambda S^\mu T^\nu dK$, *integralem habet* $u=R^{\lambda+1}S^{\mu+1}T^{\nu+1}M$, *cuius æquatio* Canonica *eſt* $dK=(\lambda+1)MSTdR+(\mu+1)MRTdS+(\nu+1)MRSdT+RSTdM$.

De-

Demonſtratio huius eadem eſt ac præcedentium: & ſimilis eſt proceſſus in infinitum.

Scholium generale.

In applicatione horum theorematum ad exempla particularia tota difficultas in eo ſita eſt, ut ex æquatione Canonica eliciatur æſtimatio indeterminatæ M. Id autem hoc modo fit: Seligo in elemento dR aliquod membrum, in quo plures dimenſiones reſpectu indeterminatarum inſunt, quam in reliquis membris, & cum hoc membro omnia membra elementi dK divido, cum quibus hæc diviſio ſuccedit, nullaque habita ratione coefficientium & ſignorum, quantitatibus per diviſionem ortis præpono coefficientes indefinitas A, B, C, D, &c. & reſultabit inde per additionem omnium, quantitas aliqua, quam voco Z, ponique poſtea debet $M=Z+N$, ſurrogando enim hanc æſtimationem indeterminatæ M in æquatione canonica & conferendo terminos æquationis reſultantis cum homologis terminis elementi dK, definientur æſtimationes litterarum aſſumtitiarum A, B, C, D.

Quod de elemento dR dictum eſt, ſimiliter intelligendum eſt de factis $Sd R$, $ST dR$, &c. Hæc vero omnia exemplis multo magis illuſtrabuntur, quam prolixioribus præceptis.

Si fuerit ſimpliciter $dK=o$, aſſumi poteſt quælibet quantitas R^λ, vel $R^\lambda S^\mu$ &c. prout hoc vel illud commodum videbitur, eſt enim etiam $R^\lambda dK$, vel $R^\lambda S^\mu dK$, &c. $=o$. & $\int R^\lambda dK$ vel $\int R^\lambda S^\mu dK$, &c. æquabitur quantitati conſtanti. Sed veniendum ad exempla particularia.

Exemplum I.

Sit ergo $du=ax^m dx$, quæritur, quid ſit u ſeu $\int ax^m dx$.

Si

Si theorema 1 ad hoc exemplum applicemus, fient $dK=adx$, $R=x$ & $\lambda=m$, & æquatio canonica $adx=(m+1)Mdx+xdM$. Quia iam in $dR=dx$, eſt unicum membrum dx, cum hoc divido $dK=adx$, oritur in quotiente a, quantitas conſtans, huius nulla habita ratione pono A, eritque adeo $Z=A$, & $M(=Z+N)=A+N$ quo valore in æquatione canonica $adx=(m+1)Mdx+xdM$ ſurrogato, oritur $adx=(m+1)Adx+(m+1)Ndx+xdN$. Comparatio terminorum homologorum præbet $A=a$: $(m+1.)$ Et $m+1\,Ndx+xdN=o$, quare poni poſſunt N & $dN=o$. Etenim ſi huius $m+1Ndx+xdN=o$, rationem habere velimus, reducetur ea per diviſionem ad $\frac{(m+1)dx}{x}+\frac{dN}{N}=o$, & integrando per *lemma* 1, $lx^{m+1}+lN=lb$, ubi b eſt conſtans, & abiectis log-mis, fiet per *lemma* 2, $Nx^{m+1}=b$, & $N=bx^{-m-1}$. Quare fiet $M=\frac{a}{m+1}+bx^{-m-1}$, & integrale quæſitum $u=Mx^{m+1}=\frac{ax^{m+1}}{m+1}+b$. Q. E. I.

Scio probe hoc exemplum nimis obvium & tritum eſſe, quam ut hoc circuitu opus fuiſſet, ſed ut methodum illuſtrarem, hæc minutim exponenda eſſe duxi.

Notandum quoque, quod, quoties poſt peractam ſubſtitutionem quantitatis $Z+N$ in locum indeterminatæ M, eiusque elementi in locum elementi huius in æquatione canonica, & poſt comparationem terminorum homologorum, ſic omnia evaneſcant, ut aliud nihil, quam æquatio $--NdR+RdN=o$ remaneat, toties poni poſſint N & $dN=o$.

Exemplum 2.

Sint $dK=mx^{m-1}ydx-nx^m dy$, $R=y$, item $\lambda=-n$

$-n-1$. adeoque $du=R^{\lambda}dK$. invenire $u=\int R^{\lambda}dK$.

In hoc exemplo æquatio canonica fit $mx^{m-1}ydx-nx^m dy=-nMdy+ydM$. Hoc loco iterum dR unico membro dy conſtat, quare divido æſtimationem ipſius dK per hoc elementum dy: in quantitate vero $dK=mx^{m-1}ydx-nx^m dy$ unicum membrum ineſt $-nx^m dy$, cum quo hæc diviſio ſuccedat, prodeunte quoto $-nx^m$, quare per monitum 1 Scholii generalis, nulla habita ratione ſigni — & coefficientis n, ſcribo Ax^m pro Z, fietque $M(=Z+N)=Ax^m+N$, & $dM=mAx^{m-1}dx+dN$. Quibus in æquatione canonica ſurrogatis, oritur $mx^{m-1}ydx-nx^m dy=mAx^{m-1}ydx-nAx^m dy-nNdy+ydN$, iam vero facta $A=1$, & deſtructis deſtruendis, manebit tantum $-nNdy+ydN=o$, quare citra omnem ſcrupulum, pono iuxta monitum ad exempl. 1. datum, $N=o$, & $dN=o$; eritque adeo $M=x^m$, & $u=R^{\lambda+1}M=x^m y^{-n}=x^m : y^n$.

Exempl. 3.

Sit $du=x^3dx : \sqrt{(aa-xx)}$, invenire $u=\int x^3dx : \sqrt{(aa-xx)}$.

In hoc exemplo fit $dK=x^3dx$, $R=aa-xx$, & $\lambda=-\frac{1}{2}$, quare æquatio canonica pro hoc exemplo, invenietur eſſe $x^3dx=-Mxdx+(aa-xx)dM$. Eſt iam iterum in dR unicum membrum $-2xdx$, cum quo divido $dK=x^3dx$, provenit $-\frac{1}{2}xx$. Quare neglecta coefficiente $-\frac{1}{2}$, ſcribo $Z=Axx$, & $M(=Z+N)=Axx+N$, & $dM=2Axdx+dN$; quibus in æquatione canonica ſuffectis, prodit $x^3dx=-3Ax^3dx+2aaAxdx-Nxdx+(aa-xx)dN$. Quare poſita $1=-3A$, id eſt $A=-\frac{1}{3}$, remanebit deſtructis deſtruendis, hæc altera $-\frac{2}{3}aaxdx-Nxdx+(aa-xx)dN=o$, vel $\frac{2}{3}aaxdx=-$

$=-Nxdx+(aa-xx)dN$. Tractetur hæc remanens æquatio, instar præcedentis æquationis canonicæ; hoc est, dividatur $\frac{2}{3}aaxdx$ per xdx, quia inest æquationi membrum $-Nxdx$, in quo æstimatio litteræ N quæritur, proveniet in quotiente quantitas constans $\frac{2}{3}aa$, quare pono $N=B+O$. & $dN=dO$, quibus in æquatione suffectis nascetur $\frac{2}{3}aaxdx=-Bxdx-Oxdx+(aa-xx)dO$, quare facta $B=-\frac{2}{3}aa$, remanebit æquatio $-Oxdx+(aa-xx)dO=o$, quare O & dO nihilo æqualia ponenda, adeo ut sit $N=B=-\frac{2}{3}aa$; hinc derivatur $M(=Z+N)=Axx+B=-\frac{1}{3}xx-\frac{2}{3}aa$. Adeoque $u(=MR^{\lambda+1})=-[(xx+2aa)V(aa-xx)]:3$.

Exemplum 4.

Sit generalius $du=x^{nm-1}dx(a+bx^m)^p$

Habemus ergo $dK=x^{nm-1}dx$, $R=a+bx^m$ & $\lambda=p$, & æquatio generalis Canonica nunc mutabitur in $x^{nm-1}dx=(p+1)bmMx^{m-1}dx+(a+bx^m)dM$, & quia in quantitate $(p+1)bmMx^{m-1}dx$ inest $x^{m-1}dx$, dividatur *quantitas* $x^{nm-1}dx$ per $x^{m-1}dx$; *quotiens* reperietur x^{nm-m}, quare facio $M=Ax^{nm-m}+N$, & $dM=nm-mAx^{nm-m-1}dx+dN$ quibus in æquat. canonica suffectis habetur $x^{nm-1}dx=(p+n)bmAx^{nm-1}dx+(nm-m)aAx^{nm-m-1}dx+(p+1)bmNx^{m-1}dx+(a+bx^m)dN$. Quare facta $A=\frac{1}{(p+n)bm}$, fiet $(m-nm)aAx^{nm-m-1}dx=(p+1)bmNx^{m-1}dx+(a+bx^m)dN$.

Fiat $N=Bx^{nm-2m}+O$, & $dN=(nm-2m)Bx^{nm-2m-1}dx+dO$, quibus in æquatione canonica secunda, quæ successit in locum primæ, surrogatis, resultat $(m-nm)aAx^{nm-m-1}dx=(p+n-1)bmBx^{nm-m-1}dx+(nm-2m)aBx^{nm-2m-1}dx+(p+1)bmOx^{m-1}dx+$

$(a+bx^m)\,d\mathrm{O}$. Quare existente $\mathrm{B}=(m-nm)\,a\mathrm{A}$: $(p+n-1)bm$ invenietur tertia æquatio canonica $(2m-nm)\,a\mathrm{B}x^{nm-2m-1}dx=(p+1)bm\mathrm{O}x^{m-1}dx+(a+bx^m)\,d\mathrm{O}$.

Ponatur tertio $\mathrm{O}=\mathrm{C}x^{nm-3m}+\mathrm{P}$, & $d\mathrm{O}=(nm-3m)\,\mathrm{C}x^{nm-3m-1}dx+d\mathrm{P}$. Quare tertia æquatio canonica mutabitur in

$(2m-nm)\,a\mathrm{B}x^{nm-2m-1}\,dx=(p+n-2)\,bm\mathrm{C}x^{nm-2m-1}dx+(nm-3m)a\mathrm{C}x^{nm-3m-1}dx+(p+1)bm\mathrm{C}\mathrm{P}x^{m-1}dx+(a+bx^m)\,d\mathrm{P}$. Idcirco facta $\mathrm{C}=(2m-nm)a\mathrm{B}:(p+n-2)bm$, invenietur quarta $(3m-nm)a\mathrm{C}x^{nm-3m-1}dx=(p+1)bm\mathrm{P}x^{m-1}dx+(a+bx^m)d\mathrm{P}$.

Simili progressu pervenietur tandem ad æquationem aliquam canonicam $(em-nm)\,a\Delta x^{nm-em-1}\,dx=(p+1)bm\,\mathrm{Z}x^{m-1}dx+(a+bx^m)d\mathrm{Z}$. In hac vero est e, terminus quicunque seriei numerorum naturalium 1, 2, 3, &c. Δ est terminus seriei coefficientum A, B, C, D cuius ordinem indicat numerus integer $e+1$. Ac denique Z est terminus seriei indeterminatarum M, N, O, P &c. cuius idem ordo est $e+1$. Sit $\mathrm{R}=a+bx^m$, & ultima hæc æquatio canonica mutabitur in æquationem $(em-nm)\,a\Delta x^{nm-em-1}dx=(p+1)\mathrm{Z}d\mathrm{R}+\mathrm{R}d\mathrm{Z}$, ducatur hæc in R^p, fietque $(em-nm)a\Delta x^{nm-em-1}dx(a+bx^m)^p=(p+1)\mathrm{R}^p\mathrm{Z}d\mathrm{R}+\mathrm{R}^{p+1}d\mathrm{Z}$, & integrando $\mathrm{R}^{p+1}\mathrm{Z}=\int(em-mn)\,a\Delta x^{nm-em-1}\,dx(a+bx^m)^p$ dicatur hoc $=\mathrm{Y}$, eritque adeo $\mathrm{Z}=\mathrm{Y}:\mathrm{R}^{p+1}$. Expansis iam omnibus fiet

$\mathrm{M}=\mathrm{A}x^{nm-m}+\mathrm{B}x^{nm-2m}+\mathrm{C}x^{nm-3m}+\&c\ldots\,\Delta x^{nm-em}+\mathrm{Y}\mathrm{R}^{-p-1}$ quare $u(=\mathrm{M}\mathrm{R}^{p+1})=(\mathrm{A}x^{nm-m}+\mathrm{B}x^{nm-2m}+\mathrm{C}x^{nm-3m}+\&c+\Delta x^{nm-em})(a+bx^m)^{p+1}+\mathrm{Y}$

ubi sunt

$\mathrm{A}=1$

$A = 1 : (p+n)bm$

$B = (m-nm)aA : (p+n-1)bm = (1-n)aA : (p+n-1)b$

$C = (2m-mn)aB : (p+n-2)bm = (2-n)aB : (p+n-2)b$

&c. &c.

$\Delta = (em-m-mn)a\Gamma ; (p+n-e+1)bm = (e-n-1) \times a\Gamma : (p+n-e+1)b$

ubi Γ eſt coefficiens, quæ ultimam Δ præcedit. Formula igitur gemino caſu integrabilis fiet

1°. Si n ſit quilibet numerus integer affirmativus, ſeu $n = e$, tunc enim fiet $Y = o$, adeoque $u = (Ax^{nm-m} + Bx^{nm-2m} + \&c. + \Delta).(a+bx^m)^{p+1}$

2°. Si $-n-p = e$ ſeu cuilibet integro affirmativo æqualis, fiet enim hoc caſu $u = (Ax^{mn+mp+m} + Bx^{mn+mp+2m} + \&c. + \Delta).(b+ax^{-m})^{p+1}$ in qua ſunt $A = -1 : mna$. $B = (1+p+n)bA : -(n+1)a$ $C = (2+p+n)bB : -(n+2)a$, &c.

Huius rei ratio eſt, quia $du = x^{mn-1}dx(a+bx^m)^p$ mutari poteſt in hanc quoque $du = x^{nm+pm-1}dx(b+ax^{-m})^p$. Sed hic poſterior integrabilitatis caſus tantum obtinet, cum p eſt integer negativus, & n integer.

Nam, quia $du = x^{qr-1}dx(b+ax^q)^p$ eiusdem formæ eſt cum hac $du = x^{mn-1}dx(a+bx^m)^p$, ea integrabilis erit per caſum priorem, ſi r fuerit numerus integer affirmativus; iam ſi in æquatione $du = x^{mn+mp-1}dx(b+ax^{-m})^p$ ponantur $m = -q$, & $n+p = -r$, mutabitur hæc æquatio in $du = x^{qr-1}dx(b+ax^q)^p$, quæ proinde integrabilis fiet, cum $r = -n-p$ æquat numerum quemcunque integrum affirmativum, hoc autem fieri nequit, niſi n fuerit numerus integer, & p quoque, ſed negativus.

3°. Item æquatio $du = x^{bm}dx(a+bx^m)^p$ integrabilis fiet, ſi p fuerit $= -e-k-\frac{1}{m}$ exiſtente e numero integro. Nam $x^{bm}dx(a+bx^m)^p = x^{bm+pm}dx(b+ax^{-m})^p$,

quare ponendo $m=-q$, & $p=-e-b+\frac{1}{q}$, mutabitur $x^{bm+pm}dx(b+ax^{-m})^p$ in $x^{eq-1}dx:(b+ax^q)^{e+b-\frac{1}{q}}$, quæ per casum primum algebraice integrabilis est. In hoc casu numerus b non est necessario integer, potest enim esse fractus quicunque

4°. Si in formula $du=x^{nm-1}dx(a+bx^m)^p$, sit $n=b+\frac{1+l}{m}$, mutabitur in sequentem $du=x^{bm+l-1}dx(a+bx^m)^p$, cuius integralis erit

$u=(Ax^{nm-m}+Bx^{nm-2m}+\&c.+\Delta x^{nm-em})(a+bx^m)^{p+1}+\Upsilon$

Iam si b fuerit cuilibet numero integro affirmativo e æqualis, fiet $em-nm=-l-1$, & $nm-em-1=l$, adeoque $Y=-(l+1)\int a\Delta x^l dx(a+bx^m)^p$. adeoque integrale est

$u=(Ax^{bm-m+l+1}+Bx^{bm-2m+l+1}+\&c.+\Delta x^{l+1})(a+bx^m)^{p+1}-(l+1)\int\Delta ax^l dx(a+bx^m)^p$. Verum $\int x^l dx(a+bx^m)^p=\int(ax^l+bx^{m+l})dx(a+bx^m)^{p-1}=\int(aax^l dx+2abx^{m+l}dx+bbx^{2m+l})dx(a+bx^m)^{p-2}=\int(a^3x^l dx+3aabx^{m+l}+3abbx^{2m+l}dx+b^3x^{3m+l})\times dx(a+bx^m)^{p-3}$ atque ita porro: unde si p fuerit integer affirmativus, mutabitur $\int x^l dx(a+bx^m)^p$ in $\int a^{p-1}x^l dx+(p-1)a^{p-2}bx^{m+l}dx+\frac{p-1\cdot p-2}{2}a^{p-3}bbx^{2m+l}dx+\&c.)\times(a+bx^m)$ quod integrale vero per præcedentia dependet a serie quantitatum absolute integrabilium $+Q\int x^l dx(a+bx^m)$ ubi Q est quantitas constans ex actuali evolutione formulæ oriunda.

Sed quid eveniet, si p sit numerus integer, sed negativus? Hæc formula $du=x^l dx(a+bx^m)^{-p}$ etiam transformari potest in

$du=(Ex^l+Fx^{m+l}+Gx^{2m+l}+\&c.)dx:(a+bx^m)$ cuius integralis dependet iterum a serie quantitatum absolute inte-

integrabilium $+Q\int x^l dx : a+bx^m$.

Hac ergo ratione huic unico exemplo inclusa sunt singula septem theoremata, quæ in Actis Erudit. 1719. sine demonstratione publicata sunt pag. 269. & theorema octavum inde etiam derivari potest. Scio quidem demonstrationes horum theorematum iam exhibitas fuisse tum a *Bülfingero* nostro, tum etiam a *Nicolao Bernoulli* V V. C C. *pag.* 467. *sq. Act. Erud.* 1720; non inconsultum tamen duxi theorema nostrum generale ad hæc ipsa applicare.

Exemplum 5.

Sit $du = 3kdz - lzzdz : zz\sqrt{(kz - lz^3 + mz^4)} = -lz^{-\frac{1}{2}}dz + 3kz^{-\frac{5}{2}}dz : \sqrt{(mz^3 - lzz + k)}$ adeoque $d\mathrm{K} = -lz^{-\frac{1}{2}}dz + 3kz^{-\frac{5}{2}}dz$, $\mathrm{R} = mz^3 - lzz + k$, & $\lambda = -\frac{1}{2}$; quare æquatio canonica nunc est $d\mathrm{K} = \frac{1}{2}\mathrm{M}d\mathrm{R} + \mathrm{R}d\mathrm{M}$. Est vero $d\mathrm{R} = 3mzzdz - 2lzdz$, dividatur ergo secundum monitum primum Scholii generalis membrum $-lz^{-\frac{1}{2}}dz$ quantitatis $d\mathrm{K}$ per membrum $-2lzdz$ quantitatis $d\mathrm{R}$, orietur in quotiene neglectis signo & coefficiente $z^{-\frac{3}{2}}$, quare pono $\mathrm{M} = \mathrm{A}z^{-\frac{3}{2}} + \mathrm{N}$; & $d\mathrm{M} = -\frac{3}{2}\mathrm{A}z^{-\frac{5}{2}}dz + d\mathrm{N}$, quibus in canonica surrogatis, prodit $-lz^{-\frac{1}{2}}dz + 3kz^{-\frac{5}{2}}dz = \frac{1}{2}l\mathrm{A}z^{-\frac{1}{2}}dz - \frac{3}{2}k\mathrm{A}z^{-\frac{5}{2}}dz + \frac{1}{2}\mathrm{N}d\mathrm{R} + \mathrm{R}d\mathrm{N}$. Quare posita $\mathrm{A} = -2$, æquatio hæc destructis destruendis præbebit $\frac{1}{2}\mathrm{N}d\mathrm{R} + \mathrm{R}d\mathrm{N} = o$. Propterea per monitum ad exempl. primum, fient N & $d\mathrm{N} = o$, atque adeo $\mathrm{M} = -2z^{-\frac{3}{2}}$ adeo-

deoque $u(=MR^{\lambda+1})=-2R^{\frac{1}{2}}z^{-\frac{3}{2}}=-2\sqrt{(mz^3-lzz+k)}:\sqrt{z^3}$.

Exemplum 6.

Sit $du=(z^5dz+z^4dz-8z^3dz):(z^5+z^4-5z^3+8z-4)$, id eſt $=(z^5dz+z^4dz-8z^3dz):(z-1)^3.(z+2)^2$. Iam cum in denominatore ſint duo factores $(z-1)^3$ & $(z+2)^2$, adhibendum eſt Theorema ſecundum, in quo iam $dK=z^5dz+z^4dz-8z^3dz$, $R=z-1$, & $S=z+2$, item $\lambda=-3$ & $\mu=-2$, itaque nunc fiet æquatio canonica $dK=-2MSdR-MRdS+RSdM$, id eſt, $z^5dz+z^4dz-8z^3dz=-(3zdz+3dz)M+(zz+z-2)dM$. Quare faciendo $M=Az^4+Bz^3+Czz$, & $dM=4Az^3dz+3Bzzdz+2Czdz$, & ſubſtitutis hiſce in æquatione canonica, fiet $z^5dz+z^4dz-8z^3dz=Az^5dz+Az^4dz-(8A+C)z^3dz-(6B+C)zzdz-4Czdz$.

Comparationes terminorum homologorum præbent $A=1$, $B=0$, & $C=0$, quare habebimus $M=z^4$. Adeoque $u(=MR^{\lambda+1}S^{\mu+1})=(z-1)^{-2}.(z+2)^{-1}z^4=z^4:z^3-3z+2$.

Exemplum 7.

Sit $du=(3q^5dx-q^4xdx+9q^3xxdx-qqx^3dx-6qx^4dx):(qq-xx)\sqrt[3]{(q^3+qqx-qxx-x^3)}$. Iam, quia $\sqrt[3]{(q^3+qqx-qxx-x^3)}=(q+x)^{\frac{2}{3}}.(q-x)^{\frac{1}{3}}$; fiet $du=(3q^5dx-q^4xdx+9q^3xxdx-qqx^3dx-6qx^4dx):(q+x)^{\frac{5}{3}}.(q-x)^{\frac{4}{3}}$. Quare nunc habemus $dK=3q^5dx-q^4xdx+9q^3xxdx-$ &c. $R=q+x$, $S=q-x$, $\lambda=-\frac{5}{3}$ $\mu=-\frac{4}{3}$, & æquationem canonicam $dK=-\frac{2}{3}MSdR-\frac{1}{3}$

$-\frac{1}{3}MRdS+RSdM$. Hoc eſt $3q^5dx-q^4xdx+9q^3xxdx$ $-qqx^3dx-6qx^4dx = (-\frac{1}{3}qdx+xdx)M+(qq-xx)dM$. Quare ſi fiat $M=A+Bx+Cxx+Dx^3$ æquatio canonica mutabitur in

$3q^5dx-q^4xdx+9q^3xxdx-qqx^3dx-6qx^4dx$
$=-\frac{1}{3}qAdx+Axdx-\frac{1}{3}qCxxdx-Cx^3dx-2Dx^4dx.$
$+qqB \quad -\frac{1}{3}qB \quad +3qqD \quad -\frac{1}{3}qD$
$\quad +2qqC$

Ex collatione terminorum homologorum eliciuntur æſtimationes coefficientum, quales ſequuntur. $D=3q$, $C=o$, $B=3q^3$ & $A=o$. Quare invenitur

$u=(3q^3x+3qx^3) : \sqrt[3]{(q^3+qqx-qxx-x^3)}$.

Exempla proxime antecedentia 5, 6, & 7um. mutuo ſumſi ex *Newtoni* tractatu de *Quadraturis*, quibus Celeberr. Auctor ſeriem ſuam Theoremate III. pro quadraturis exhibitam, illuſtravit. Sed quid vetat, quominus hanc ſeriem ipſam ex primo noſtro theoremate derivemus? Eſt ergo

Exemplum 8.

Sint $R=e+fz^q+gz^{2q}+hz^{3q}+$ &c. dein $Z=a+bz^q+cz^{2q}+dz^{3q}+$ &c. & $du=R^{l-1}Z^{m-1}dz$. Invenire integralem huius formulæ.

Eſt itaque $dK=Z^{m-1}dz$, $\lambda=l-1$, & per theor. 1. æquatio canonica $Zz^{m-1}dz=lMdR+RdM$. Ponamus iuxta præcedentia $M=Az^m+Bz^{m+q}+Cz^{m+2q}+Dz^{m+3q}+$&c.$+N$, eritque

$RdM=meAz^{m-1}dz+(m+q).eBz^{m+q-1}dz+(m+2q)eCz^{m+2q-1}dz+$ &c. RdN
$\quad +mfA \quad +(m+q).fB$
$\quad +\quad m.gA$
$lNdR= \quad +flqAz^{m+q-1}dz+2glqAz^{m+2q-1}dz+$&c.$+lNdR$
$\quad +flqB$

Ergo æquatio canonica mutatur in ſequentem :

$$az^{m-1}dz+bz^{m+q-1}dz+cz^{m+2q-1}dz+\&c. =$$
$$meAz^{m-1}dz+(lq+m)fAz^{m+q-1}dz+(2lq+m)gAz^{m+2q-1}dz+lNdR+RdN.$$
$$+(m+q)eB \qquad +(lq+m+q)fB$$
$$+(m+2q)eC$$

Iam comparatio terminorum homologorum præbet :

$$A=\{a - - -

est h. In aliis casibus omnibus æstimationes litterarum G, H, I, &c. facili negotio obtinebuntur.

Si series $A+Bz^q+Cz^{2q}+Dz^{3q}+$ &c. alicubi abrumpens est, Y fiet $=0$, adeoque formula absolute integrabilis erit.

Non absimili modo ex theoremate nostro secundo derivare licet Theorema IV. *Newtoni* Propositione VI^ta *Tract. de Quadraturis* exhibitum, sed quia hæc deductio prolixiorem deposcit calculum, ab ea nunc abstinebo.

Pergam ergo ad illustrationem theorematum nostrorum per exempla quantitatum differentialium, duas aut plures indeterminatas varie inter se mixtas, vel subinde secunda differentialia involventium, qualia non mihi ipsi ad arbitrium finxi, sed ab amico mihi sine integralibus suppeditata vel aliunde desumta sunt. Ad talia enim perinde sese extendit ac ad ea, quæ in superioribus tractavi. Cuius rei leve specimen exemplo secundo superiori iam exhibui.

Exemplum 9.

Sit $du=3a^3yydy-6aaxxydy+3ax^4dy-6aaxyydx+12ax^3ydx-6x^5dx$.

Hæc æquatio æquivalet alteri $du=(3a^3x^{-5}yydy-6aax^{-3}ydy+3ax^{-1}dy-6aax^{-4}yydx+12ax^{-2}ydx-6dx)x^5$. Quo casu fit $dK=3a^3x^{-5}yydy-6aax^{-3}ydy+3ax^{-1}dy-6aax^{-4}yydx+12ax^{-2}ydx-6dx$, $R=x$, $\lambda=5$, & æquatio canonica $dK=6Mdx+xdM$. Iam ut inveniri possit, quid sit M, in quantitate dK divido per dx omnia illa membra in quibus dx inest, prodibunt neglectis coefficientibus & signis $x^{-4}yy$, $x^{-2}y$ & 1, quare pono $M=Ax^{-4}yy+Bx^{-2}y+C+N$. Hoc autem valore litteræ M, eiusque elementi dM, in canonica substitutis proveniet æquatio:

$+3a^3x^{-5}yydy-6aax^{-3}ydy+3ax^{-1}dy-6aax^{-4}yydx$
$+12ax^{-2}ydx-6dx$
$=2Ax^{-3}ydy+Bx^{-1}dy+2Ax^{-4}yydx+4Bx^{-2}ydx$
$+6Cdx+6Ndx+xdN$

Comparatio terminorum homologorum præbet, $A=-3aa$, $B=3a$, $C=-1$, & remanebit æquatio sequens

$$3a^3x^{-5}yydy=6Ndx+xdN.$$

Ducatur hæc æquatio in x^5, provenietque $3a^3yydy=6Nx^5dx+x^6dN$. Utraque pars huius æquationis est integrabilis, invenietur enim integralis eius $a^3y^3=Nx^6$. Habemus ergo $N=a^3x^{-6}y^3$, adeoque $M(=Ax^{-4}yy+Bx^{-2}y+C+N)=-3aax^{-4}yy+3ax^{-2}y-1+a^3x^{-6}y^3$, adeoque

$u(=MR^6=Mx^6)=-3aaxxyy+3ax^4y-x^6+a^3y^3$
$=a^3y^3-3aaxxyy+3ax^4y-x^6=(ay-xx)^3$.

Hoc idem integrale inveniſſemus ponendo $dK=3ayydy-6aaxxydy+3ax^4dy-6aaxyydx+12ax^3ydx-6x^5dx$. Dein $R=x$ vel $=y$, ſed $\lambda=o$, adeoque $\lambda+1=1$.

Exemplum 10.

Sit $du=(-\frac{3}{2}cyydy+\frac{3}{10}abbty^{-1}dy)\sqrt[4]{(a^3b-2cy^3+3abbt)^3}$ exiſtente $t=\sqrt[5]{accyy}$. Hoc loco eſt $dK=-\frac{3}{2}cyydy+\frac{3}{10}abbty^{-1}dy$, $R=a^3b-2cy^3+3abbt$, $\lambda=-\frac{3}{4}$, adeoque $\lambda+1=\frac{1}{4}$. & æquatio canonica $dK=\frac{1}{4}MdR+RdM$. Nunc vero, quia $\frac{1}{4}dR=dK$, erit $M=1$, adeoque $u(=MR^{\frac{1}{4}})=\sqrt[4]{(a^3b-2cy^3+3abbt)}$.

Exemplum 11.

Si $du=(2axyydx+cxyydy-cy^3dx-2ccyydy-4axxydx+4ax^3dy+8ccyydx-2axxydy):(y^4-8xy^3+16xxyy)$.

Quoniam $y^4-8xy^3+16xxyy$ eſt quadratum, cuius radix eſt

eſt $yy-4xy$; fiat $R=yy-4xy$, $\lambda=-2$, & $dK=2axyydx+cxyydy-cy^3dx-2ccyydy-4axxydx+$ &c. & æquatio canonica fiet, $dK=-MdR+RdM$. Eſt vero $dR=2ydy-4ydx-4xdy$. Reſpicio nunc ad omnia membra quantitatis dK, quæ per ydx dividi poſſunt, & præſcindendo a coefficientibus & ſignis, inveniuntur quoti xy, yy, xx, y. Pono ideo

$M=Ay+Bxy+Cxx+Dyy$, & inveniemus æquationem canonicam mutari in æquationem

$+2axyydx+cxyydy-cy^3dx-2ccyydy-4axxydy$
$+4ax^3dy+8ccyydx-2axxydy=+2Cxxydx-Bxyydy$
$+4D$
$+By^3dx-Ayydy-4Cxxydy+4Cxxdy+4Ayydx$
$+6D$
$-2Cxxydy-2Dy^3dy$

Ex comparatione terminorum elicientur $A=2cc$, $B=-c$, $C=a$, & $D=o$, quare fit $M=2ccy-cxy+axx$, adeoque $u(=MR^{-1})=(2ccy-cxy+axx):(yy-4xy)$.

Exemplum 12.

Si $du=(4xyzdy-xxzdx-x^3dz+2xy^2zdx+2xyzdx-2y^2zdy-2y^2zdy-2y^3dz-x^2zdy+xxydz):(xx-2xy+yy)$.

Erunt $dK=4xyzdy-x^2zdx-x^3dz+2xy^2dz+2xyzdx-2y^2zdy-2y^2zdx-2y^3dz-xz^2dy+x^2ydz$; $R=x-y$, $\lambda=-2$ adeoque æquatio canonica fit $dK=-MdR+RdM$. Iuxta ſuperiora invenietur ſumendum eſſe $M=Ax^2z+Bxyz+Cy^2z$, ope huius mutabitur canonica in æquationem cuius termini homologi inter ſe comparati præbebunt $A=-1$, $B=o$, & $C=2$, adeoque $M=-xxz+2yyz$, & $u(=MR^{-1})=2yyz-xxz:x-y$.

Exemplum 13.

Si $-addx = (ydx-xdy)^2 x : (xx+yy)^{\frac{3}{2}}$, in hypothesi quod $ydx-xdy$ sit constans. Integrabitur ponendo $du = -addx : ydx-xxdy$, $R = xx+yy$, $\lambda = -\frac{3}{2}$, & $\lambda+1 = -\frac{1}{2}$ adeoque æquatio canonica pro hoc exemplo invenietur esse $dK = -\frac{1}{2}MdR+RdM$, atque adeo $xydx-xxdx = (-xdx-ydy)M+(xx+yy)dM$; Fiat $M = Ay$, & mutabitur æquatio in $xydy-xxdy = -Axydx+Axxdy$. Quare $A = -1$. & $u(=MR^{-\frac{1}{2}}) = -y : V(xx+yy)$.

Atqui propter differentiale constans $ydx-xdy$, erit etiam $u = -adx : ydx-xdy$ quare $\frac{-adx}{ydx-xdy} = \frac{-y}{V(xx-yy)}$; hinc $adx = \frac{yydx-xydy}{V(xx+yy)}$, vel $abdx = \frac{byydx-bxydy}{V(xx+yy)}$ ducatur in x^{-2} fiet $-abx^{-2}dx = (bx^{-1}ydy-bx^{-2}yydx)) : V(xx+yy)$; si nunc iterum sit $R = xx+yy$ fiet pro hac æquatione competens æquatio canonica $bx^{-1}ydy-bx^{-2}yydx = (ydy+xdx)M+(xx+yy)dM$; quare si ponatur $M = bx^{-1}$, æquatio hæc fiet identica; est ergo $c+\frac{ab}{x}$, integrale ipsius $-abx^{-2}dx = MR^{\frac{1}{2}}$, id est $= \frac{bV(xx+yy)}{x}$. Quæ æquatio est ad sectiones conicas.

Exemplum 14.

Si $fdx = (dxdy^3+dx^3dy-xdxdyddx+xdx^2ddy) : 2x^3dy^3$. Integrare æquationem. Pone $dK = \frac{1}{2}dx+\frac{1}{2}dx^3dy^{-2}-\frac{1}{2}xdxdy^{-2}ddx+\frac{1}{2}xdx^2dy^{-3}ddy$, $R = x$, & $\lambda = -3$, adeoque $\lambda+1 = -2$. Quare æquatio canonica nunc fit $dK = -2Mdx+xdM$. Elicietur ergo $M = A+Bdx^2ay^{-2}$. Ope huius vero æquatio canonica abit in

$\frac{1}{2}dx$

$\frac{1}{2}dx + \frac{1}{2}dx^3\,dy^{-2} - \frac{1}{2}xdxdy^{-2}\,ddx + \frac{1}{2}xdx^2\,dy^{-3}\,ddy$
$= -2Adx - 2Bxdxdy^{-2}\,ddx - 2Bxdx^2\,dy^{-3}\,ddy$, quare $A = -\frac{1}{4}$,
$B = -\frac{1}{4}$ & $\int\!\!\int dx = -\frac{1}{4xx} - \frac{dx^2}{4xxdy^2}$, vel addita constanti n.
$\int\!\!\int dx = (4nxxdy^2 - dx^2 - dy^2) : 4xxdy^2$.

Videantur ad hæc duo postrema exempla, *les Memoires de l'Acad. Royale des Sciences* 1710. circa finem.

DE INTEGRATIONIBVS AEQVATIONVM DIFFERENTIALIVM,

ubi traditur

METHODI ALICVIVS SPECIMEN INTEGRANDI

sine prævia separatione indeterminatarum:

Auctore Joh. Bernoulli.

I.

QUando æquatio aliqua differentialis primi gradus reducta habetur ad $pdx = qdy$, ubi p datur per x, & q per y, hoc est, meo loquendi more, ubi p & q sunt functiones qualescunque datæ indeterminatarum x & y; eo casu constructio æquationis nulla premitur difficultate, concessis nimirum quadraturis; quæ & ipsæ generaliter ad extensiones curvarum algebraicarum non ita pridem reductæ sunt, vid. Cel. Hermanni schediasma in actis Lipsi. 1723. meumque de hac materia analytice tractanda editum in iisdem actis 1724. Adeo ut hac in parte nihil ulterius ad maiorem perfectionem desi-

M. Iun. 1726.